

GAURAV ALOK GUPTA

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EDUCATION

MSc., COMPUTER SCIENCE (Average Grade: 16.45/20)

University of St Andrews, Fife, Scotland, United Kingdom | September 2025 – August 2026 (Expected)

- Modules: Distributed Systems, Concurrency & Multi-core Architecture, Data-Intensive Systems, Software Architecture & Design, Critical Systems Engineering.

B.Tech., COMPUTER SCIENCE AND TECHNOLOGY (CGPA: 8.88/10)

Dayananda Sagar University, Bengaluru, Karnataka, India | August 2019 – May 2023

PROFESSIONAL EXPERIENCE

TALLY SOLUTIONS PVT. LTD. - BENGALURU, INDIA | May 2023 - June 2025

Software Development Engineer - II | C++, Python, AWS, OCI

- Engineered an **automated failover and self-healing system** in Tally Prime on AWS, restoring operations for **30+ client systems** during outages. Recognised with the **SPOT Award as Customer Delight Champion**.
- Developed **20+ custom Lambda functions** across **10+ components** to enable **zero-loss cloud platform migration** between AWS and OCI, eliminating vendor lock-in and improving scalability.
- Improved core product stability by resolving critical defects in **XML and Math Parsers** and the **Tally-to-Tally migration tool**, preventing data loss for enterprise customers.
- Delivered key product features - **file load source identification** and **placeholder collection attributes** - in TallyPrime Developer, strengthening third-party TDL integration.

PROJECTS

INTERVIEW PREPARATION AGENT | Python, LangGraph, Gemini API, Tavily API | [GitHub](#)

- Built a stateful **LangGraph agent** with three modes — CS/networking revision **dynamically adapts question difficulty using per-topic confidence scores** persisted in SQLite, behavioural coaching uses **structured LLM extraction to score and rewrite answers** in STAR format, tracking competency coverage across sessions. Question banks self-update via **Tavily web search** with deduplication.
- Designed a **CI pipeline** where an **LLM-powered code reviewer runs on every push and PR**, posts inline review comments with severity-graded findings on pull requests, auto-fixes critical issues on pushes to main, with **retry/backoff for rate-limit resilience**. **115 unit tests with full LLM mocking** run on every commit.

GENIESPEAKS: PRODUCT & REVIEW AGGREGATOR | Python, ReactJS, MongoDB, NodeJS | [GitHub](#)

- Designed and built a **full-stack e-commerce aggregator** with **scheduled web scraping** of Amazon and Flipkart, incorporating **NLP-based sentiment analysis** on customer reviews.

SKILLS

LANGUAGES AND FRAMEWORKS | C++, Python, JavaScript TypeScript,, Haskell, React, FastAPI HTML, CSS, Sass

CLOUD AND DEVOPS | AWS (Lambda, EC2, S3), Oracle Cloud Infrastructure (OCI), Docker

AI AND INTELLIGENT SYSTEMS | Gemini API, LangChain/LangGraph, LLM Integration

DATABASE TECHNOLOGIES | SQL, MongoDB, S3

TOOLS AND PLATFORMS | Git, GitHub, Jira, Confluence, Figma, Postman

EXTRACURRICULARS AND VOLUNTEERING

- Process, Methodology, and Tools (PMT) trainer and Hackathon Mentor for 10 teams (2024). Tally Solutions Pvt. Ltd.
- Shortlisted for JPMC Code for Good Hackathon (2022). JP Morgan Chase & Co.
- C Language Trainer for 70+ students and pre-placement session lead. DSU (University)